

Maaz Qureshi

Research Engineer (Embodied AI and Robotics): MASc Mechtronics University of Waterloo

[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | +1 437 870 8791 | m23qures@uwaterloo.ca | Waterloo, Canada

SKILLS

Machine Learning & Foundation Models: Vision-Language-Action models (π_0 , $\pi_{0.5}$), World-Action Models (literature), Parameter-Efficient Fine-Tuning (PEFT: LoRA), CNNs, VLMs, Quantization (FP16/INT8), Reinforcement Learning (PPO, policy optimization), Imitation Learning (behavior cloning) & Diffusion Policy

Robotics: 6D Pose Estimation, Visual-Inertial SLAM (volumetric mapping & localization), Motion Planning for 7-DoF Manipulators (IK/FK), Sensor Fusion (stereo RGB-D, LiDAR, mmWave 4D radar, IMU), 5G-Connected Factory Robotics, AR/XR (ARKit)

Languages: Python, C++, PyTorch, TensorRT (CUDA), Eigen, NumPy, ROS2

Tools: Docker, CI/CD, gRPC, NVIDIA Isaac, MuJoCo, Gazebo, RViz, Asynchronous GPU Pipeline, Debugging/Monitoring (Valgrind, GDB, Jtop)

RESEARCH EXPERIENCE

Embodied AI Robotics Research Assistant

May 2026 – Present

[Apple Inc.](#) (joint research with University of Waterloo)

Waterloo, Canada

- Confidential NDA project with Apple Robotics and University of Waterloo for 7-DoF manipulator motion planning.
- Built an iPhone Pro (ARKit)-based egocentric human-demonstration data pipeline for PEFT (LoRA-based) fine-tuning of foundation models.
- Fine-tuning the π -family model for Apple factory tasks on the Franka Panda arm; a world-action model pipeline is in progress (literature).
- Characterized ARKit pose-tracking accuracy against Franka Panda 6D pose and gripper-width ground truth across a 5–65°/s angular-velocity sweep, identifying a sub-40°/s operating envelope (mean translational error <10%) for reliable demonstration collection.
- Developing a UMI-style handheld gripper with a custom Franka-flange mount for scalable demonstration collection and deployment.

Robotics Software Engineer [\[video\]](#)

September 2025 – April 2026

[Cobionix Corporation Inc.](#)

Kitchener, Canada

- Developed a 22 FPS real-time multimodal RGB-D ML pipeline (SmolVLM, pose estimation, object detection, Segmentation) in C++/CUDA.
- Applied PEFT (LoRA) to fine-tune the SmolVLM vision-language backbone for object recognition.
- Applied TensorRT FP16/INT8 quantization in async GPU pipelines on NVIDIA Jetson to boost throughput and cut latency.
- Implemented real-time RGB-D stereo sensor fusion with intrinsic/extrinsic spatiotemporal calibration (>90% alignment accuracy).

Graduate Research Student: ML Robot Perception & Manipulation

January 2024 – August 2025

[University of Waterloo, RoboHub Lab](#) [\[Publication\]](#), [Video](#), [GitHub](#)

Waterloo, Canada

- Developed a distributed multi-robot dense volumetric SLAM system (C++/ROS2 Humble) with TSDF-octree mapping and ORB-SLAM3.
- Achieved an 85% improvement in map coverage over the state-of-the-art baseline across 4 autonomous mobile robots (AMRs).
- Fused mmWave 4D radar with RGB-D sensing for perception in cluttered, low-visibility settings (92% detection accuracy).

MITACS Industrial Research Internships:

[Keysight Technologies Inc.](#) [\[Publication\]](#), [Video](#), [GitHub](#)

Thesis Internship

- (SIM-to-REAL) Developed an OMPL/RRT* sampling based motion-planning pipeline for 7-DoF Franka and KUKA collaborative robots.
- Benchmarked a PPO reinforcement-learning planner against RRT sampling-based motion planning for manipulator trajectories.
- Validated sim-to-real deployment, achieving 100% joint-limit and collision avoidance across hemispherical scan patterns.
- Built with Python, ROS/RViz, and Docker for end-to-end pipeline development and deployment.

[Rogers Communications Inc.](#)

Thesis Internship

- Integrated 5G connectivity into a distributed software pipeline on NVIDIA GPUs for smart-factory robots, achieving a 37% latency reduction.
- Benchmarked 5G sub-6 GHz (3.5 GHz) and mmWave (28 GHz) channels against WiFi for industrial robotics applications.

Asset Integrity Engineer: Robotics & ML

May 2022 – December 2023

[ABYSS Solutions Inc.](#)

Islamabad, Pakistan (Onsite) & Remote Australia

- Performed anomaly outlier detection (supervised ML, U-Net segmentation) for corrosion detection on oil/gas platforms.
- Performed offline SLAM mapping visual model for mooring-chain and dam inspection using underwater ROV imagery.[\[Link\]](#)
- Led a 10-engineer team delivering end-to-end digital inspection of 18,000+ assets (deadlegs inspection) with 91% high-quality coverage.
- Actively involved in special projects of organization for UAV, Virtual reality and Underwater robotics ops.

EDUCATION

University of Waterloo

Waterloo, Canada

Master of Applied Science (MASC), Mechatronics Engineering (CGPA: 93.5%)

January 2024 – August 2025

- Thesis link: "Intelligent Multi-Robot Autonomy with Connected AMRs and Manipulators for SMART Factory(s)", Transcript, Degree
- Advisor(s): Dr. William Melek and Dr. George Shaker . Defence committee: Dr. Soo Jeon, and Dr. Gennaro Notomista

Bahria University

Islamabad, Pakistan

Bachelor of Science (BSc), Electrical (Electronics) Engineering (CGPA: 80.5%)

September 2017 – August 2021

- Thesis link: "Path Planning and Perception for multiple mobile robot Swarm Robotics", Transcript, Degree
- Advisor: Dr. Saleem Aslam

SIDE PROJECTS

- Diffusion Policy vs Imitation learning in Navigation:** [Link] benchmarked diffusion-policy, behavior-cloning, and LSTM imitation learning for TurtleBot 4 navigation; added YOLOv11 for real-time safety detection.
- Visual Feature Extraction with Machine Learning:** Applied KNN and SVM regression models with FRIQUE image-quality metrics to benchmark visual feature extraction on computer-generated vs. real-world image datasets.
- Humanoid Robotics: Non-Verbal Human-Robot Interaction** [Link]: Designed a non-verbal gesture-recognition model for multimodal control of a NAO humanoid robot assisting in collaborative breakfast preparation.
- AMRs in Health Care** [Link]: Built an RFID-based autonomous nursing robot for pill dispensing, real-time perception, and live video streaming in healthcare environments.
- UAS Challenge 2022, IMechE UK: Aerial Robotics** [Link]: Team Supervisor for a hexacopter UAV (6.9 kg takeoff weight, 100 m operating altitude) for locust-swarm spraying, with a custom carbon-fiber S950 frame and 340 KV BLDC motors delivering 2920 g of thrust.

SELECTED PUBLICATIONS:

- [C1] M. Qureshi et al., "CRADMap: Applied Distributed Volumetric Mapping with 5G-Connected Multi-Robots and 4D Radar Perception," *IEEE ICARM 2025*. (Published) [IEEE Link]
- [C2] M. Qureshi et al., "Hemispherical Angular Power Mapping of Installed mmWave Radar Modules Under Realistic Deployment Constraints," *IEEE APS 2026*. (Published) [arXiv]
- [J1] M. Qureshi et al., "RAPTAR: Radar Radiation Pattern Acquisition through Automated Collaborative Robotics," *IEEE Robotics and Automation Letters (RA-L)*. (submission under revision) [arXiv]

AWARDS

- Full Graduate Funding Award, MASC, University of Waterloo.
- Top 3 Undergraduate Thesis Project, Bahria University.
- Best Performer, Employee of the Month, Abyss Solutions Inc.
- 3× Hackathon Winner, Abyss Solutions Robotics & VR Innovation Challenges.

OTHER EXPERIENCE

TA, University of Waterloo

Spring 2024 – Spring 2025

ME 123 (Electrical Eng. for Mechanical Engineers), Spring 2024 | ME 360 (Automatic Control Systems), Fall 2024 & Winter 2025 | MTE (Advanced Calculus), Spring 2025

Conference Presentation, IEEE APS 2026

Detroit, Michigan, USA

Jul. 2026

Conference Presentation, IEEE ICARM 2025

Portsmouth, United Kingdom

Aug. 2025

Symposium Presentation: RoboHub Symposium, Poster & Research Talk

University of Waterloo, Canada

Jan. 2025

Conference Presentation, Rogers "Beyond Connectivity" Summit

Brampton, Canada

Jul. 2024

PEER REVIEW SERVICE

- Reviewer, IEEE International Conference on Robotics and Automation (ICRA) 2026 (Vienna, Austria; Submission #1761): "ROR-SLAM: UWB Radar-based Object Recognition for Feature-based SLAM."
- Reviewer, IEEE Sensors Journal (Manuscript Sensors-69555-2024): "EAMT-SLAM: Edge-Aided Monocular Thermal Sensor SLAM in Low-Illumination Environments."
- Reviewer, MEIE 2026 (9th Int'l Conference on Mechanical, Electric, and Industrial Engineering; Manuscript MEIE99621): "Feasibility Analysis and Reinforcement Design of a Portable Radar for Use in Vehicle Environments." Recommended minor revisions.